Robot Navigation Anomaly Detection and Causal Analysis

Team 02 Final Presentation

AI Algorithms – Theory and Engineering
Winter Semester 2025/26

January 27, 2026

Outline

Requirements Identified

- Metrics to quantify localization and navigation performance (For anomaly detection)
- Anomaly detection (Single source and Multiple source)
- Feature vectors for Anomaly Prediction
- Anomaly Prediction FOL Rule Derivation
- Scenario based and log based predictions
- Implement as Python script (*.py) or package that includes the complete pipeline

Dataset Overview

Dataset Properties

- **Total Scenarios:** 100 directories
- **Runs per Scenario:** 3 (labeled 0, 1, 2)
- **Total Runs:** ∼300 experiments
- **Robot:** TurtleBot4 (0.22m diameter)

Environment Types

- door-width
- door-size
- room-size
- hallway-window

File Structure per Run

File	Content
poses.csv	Robot trajectories
behaviors.csv	BT execution logs
rosbag2.csv	ROS2 bag + AMCL
scenario.config	Goals, params
run.yaml	Run metadata

Pre-processing Steps

- **Data Synchronization**: Aligns the robot's estimated position with the "Ground Truth" data.
- **Stationary Filtering**: Removes irrelevant data where the robot was not moving, typically at the start or end of a run.
- **Coordinate & Orientation Normalization**:
- **Yaw Normalization**: Normalizes all yaw angles to the range $[-\pi, \pi]$.
- **Quaternion Conversion**: Converts orientation data from quaternion (w, x, y, z) to yaw angles for easier error calculation.

Performance Metrics (8 Metrics)

Metric	Unit	Description	Category
mean_pos_error	m	Average position error	Localization
rmse_pos	m	Root mean square error (spike sensitive)	Localization
mean_yaw_error	rad	Average orientation error	Localization
executed_path_length	m	Total path traveled	Path
path_efficiency	ratio	gt_path / executed (1 is best)	Path
mean_linear_velocity	m/s	Average movement speed	Kinematics
trajectory_smoothness	rad/s^2	Angular acceleration (lower = smoother)	Kinematics
duration	s	Total run time	Kinematics
mean_amcl_uncertainty	m	Average localization confidence	AMCL

Features for Causal Analysis (27 Features)

Anomaly Type	Primary Predictors	Secondary Predictors
High AMCL Unc.	high_noise, noise_level	—
Stuck	in_narrow_corridor, in_small_room, tight_obstacle_clearance	corridor_width, room_area
High Yaw Error	high_noise, noise_level, in_small_room	total_obstacle_area, near_wall
Pos. Error Spike	high_noise, noise_level, in_small_room	total_obstacle_area, near_wall
Path Inefficiency	obstacle_clearance_ratio, num_obstacles, total_obstacle_area	min_door_narrow, goal_through_door
Goal Failure	goal_near_wall, waypoint_in_tight_space, door_too_narrow	goal_wall_distance, door_width
Oscillation	in_narrow_corridor, in_small_room, tight_obstacle_clearance	corridor_width, room_area

Anomaly Detection and Classification

Rule-Based Anomalies (8 Types)

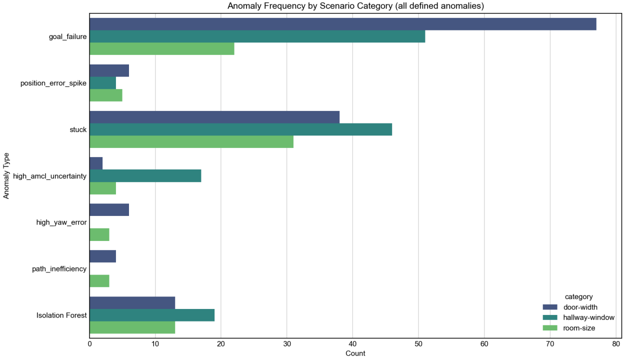
- **goal_failure**: Goal not reached
- **no_initiation**: Robot didn't start
- **position_error_spike**: Sudden errors
- **stuck**: Robot stopped moving
- **high_amcl_uncertainty**: Poor localization
- **high_yaw_error**: Orientation issues
- **path_inefficiency**: Sub-optimal path
- **oscillation**: Jerky motion (smoothness > 2.0)

ML-Based Anomaly Detection

Isolation Forest trained on 8 standardized metrics:

- Captures multi-metric abnormal patterns missed by rules
- Unsupervised learning for outlier detection

Anomaly Detection Results



Key Findings

- Total occurrences: **341**
- **goal_failure** is most common (50%)
- **stuck** follows at 38.3%
- **Isolation Forest** catches 15%
- Single run can have multiple anomalies

Anomaly Correlations Analysis

- **position_error_spike + high_yaw_error** (Jaccard = 0.60)
  $\rightarrow$ Poor orientation tracking leads to position drift
- **high_yaw_error + path_inefficiency** (Jaccard = 0.78)
  $\rightarrow$ Orientation errors cause inefficient path execution

Early Prediction: Predictability

Anomaly	PR-AUC	ROC-AUC	Cases
goal_failure	0.852	0.828	150
stuck	0.766	0.789	115
Isolation Forest	0.661	0.834	45

Early Multi-features and Anomaly Correlation

Stuck

- path_length: r=-0.491
- std_velocity: r=-0.437
- path_efficiency: r=-0.384

Goal Failure

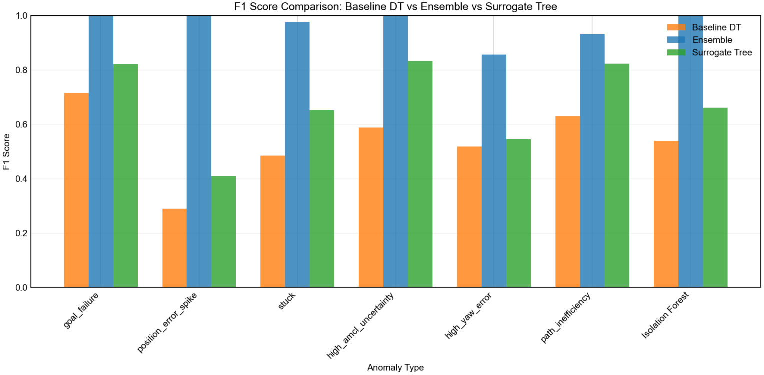
- duration: r=+0.524
- mean_angular_vel: r=-0.283
- path_length: r=+0.276

high_amcl_uncertainty

- rmse_pos: r=+0.291
- max_velocity: r=+0.277
- std_pos_error: r=+0.274

Causal Analysis Methodology Evolution

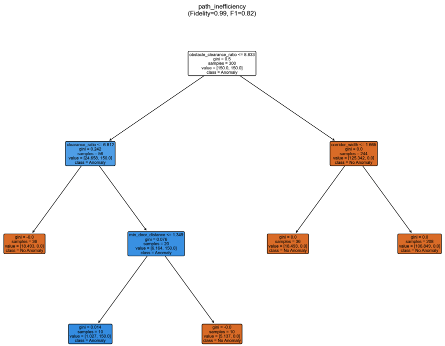
Simple Decision Tree → Ensemble Model → Surrogate Model



Causal Analysis Current Approach

- **Step 1: High-Capacity Model Training (Ensemble)**
  - → Voting Classifier: Decision Tree + Random Forest + Gradient Boosting
  - → Separate trees for both log-based and scenario-based predictions
  - → Prioritizes F1-score
- **Step 2: Surrogate Model Training (Distilled Decision Tree)**
  - → Separate trees for both log-based and scenario-based predictions
  - → Constrained tree (max_depth=4)
  - → Target: Ensemble predictions (not ground truth)
- **Step 3: Rule Extraction**
  - → Extract paths with P(Anomaly) > 0.5
- **Step 4: FOL Translation**
  - → Map thresholds to semantic predicates

What Are Tree Models? and Why?



Why Surrogate? Captures ensemble's knowledge in an interpretable tree structure, enabling quality FOL rule extraction.

Technical & Methodological Insights

1. **Better quality rules with just scenario information**
   - In case of many anomalies it showed better performance (F1 score, Precision, Recall, other metrics) when just used scenario description, robot information, and environment JSON file for deriving FOL rules instead of also including more informative logs, CSV files.
2. **Surrogate Distillation**
   - Ensemble models achieve higher accuracy, but surrogate trees provide 90%+ fidelity with full interpretability.
   - Surrogate trees were even simpler than simple decision trees yet showed better results (F1, confusion metrics).
3. **Statistical Caution**
   - Be cautious with "perfect" confidence and performance (F1) on tiny supports.

Domain & Navigation Insights

1. **Geometry Dominates**: Obstacle clearance ratios, door distance/width, wall distances explain most failures
2. **Corridors are High-Risk**: Corridors represent a high-risk context for goal failure (log-based evidence)
3. **Hazard Zones**: Narrow/near-door situations are recurring hazard zones across multiple failure modes

Team 2 Final Presentation

Thank you!

Questions?